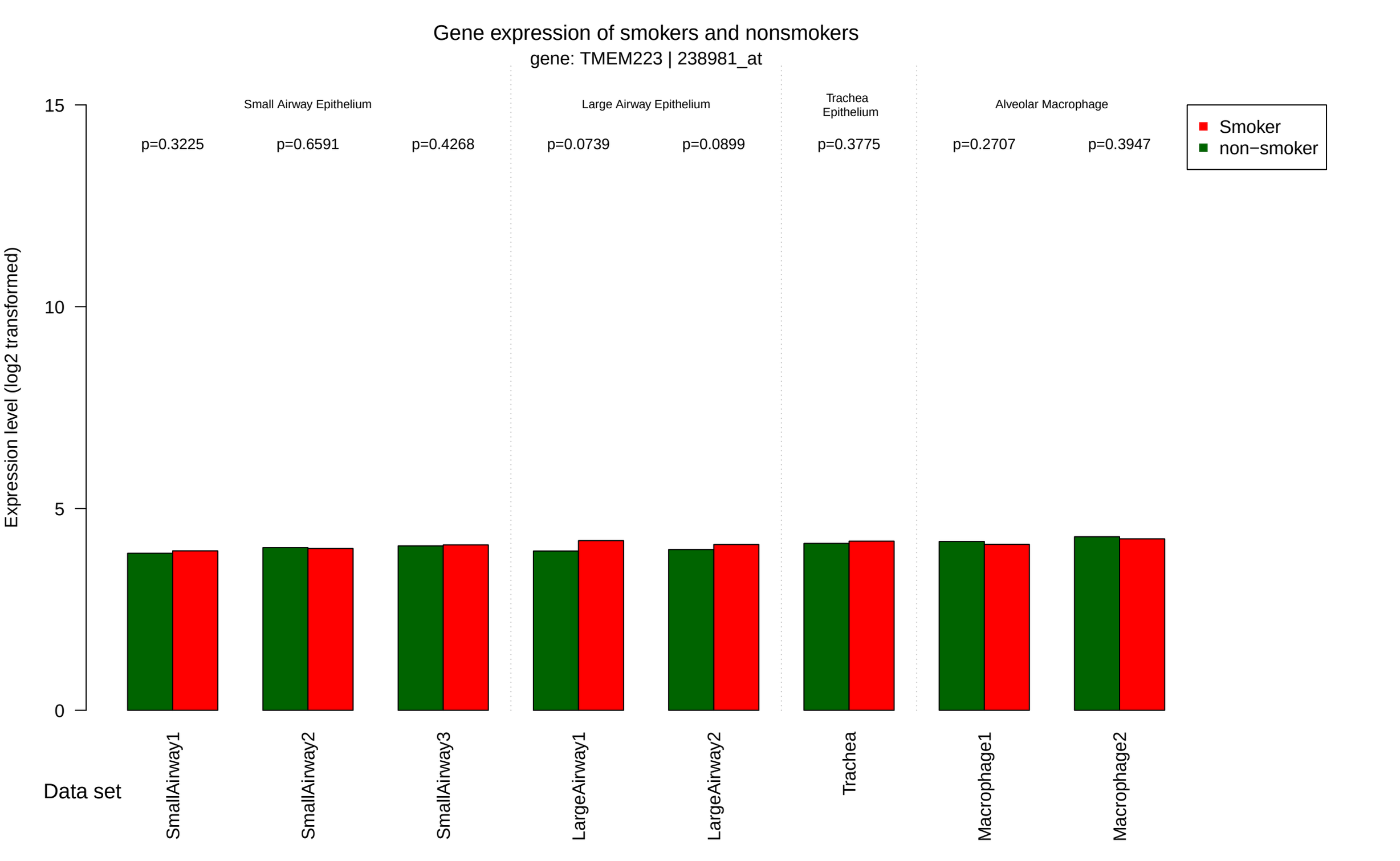
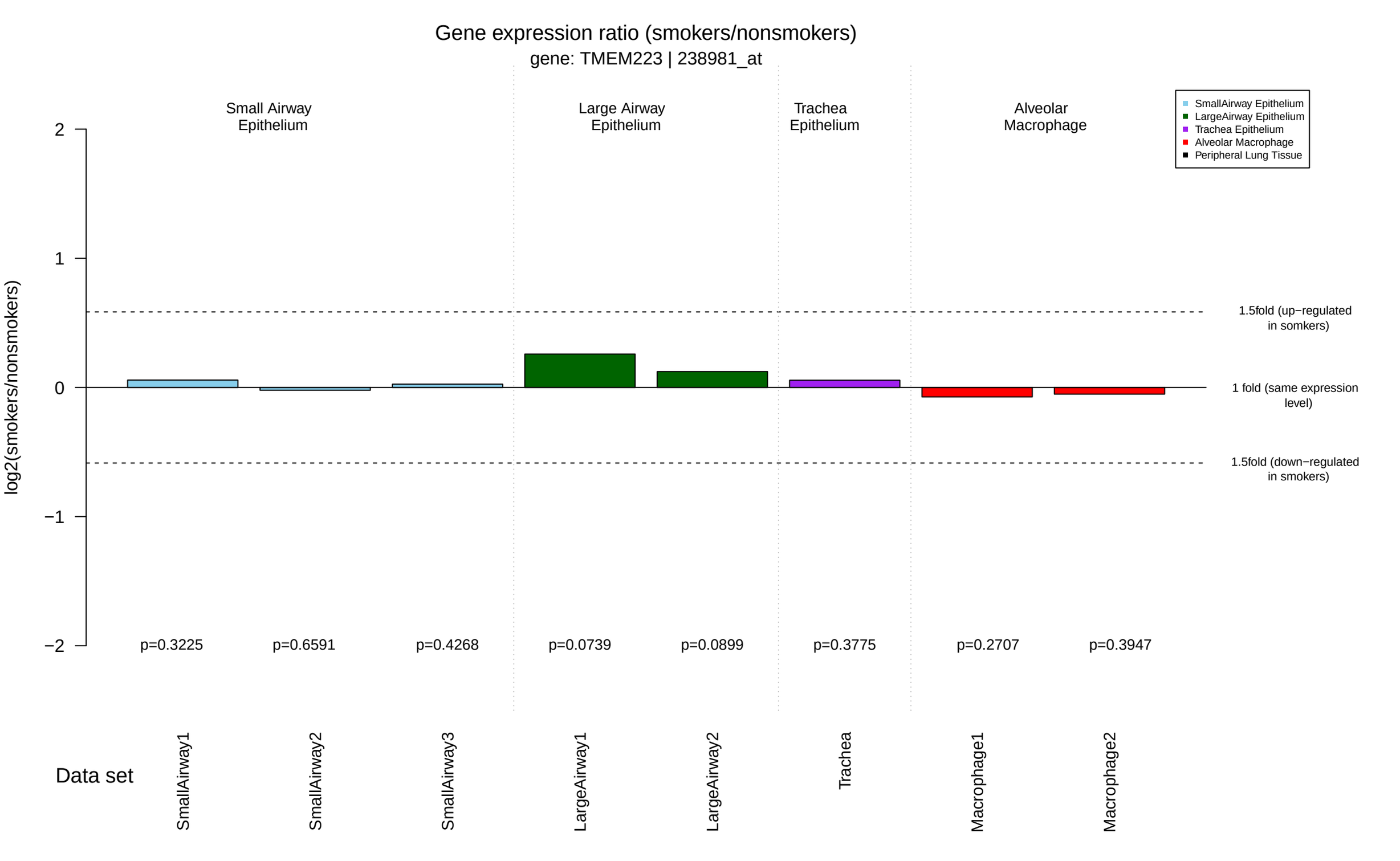
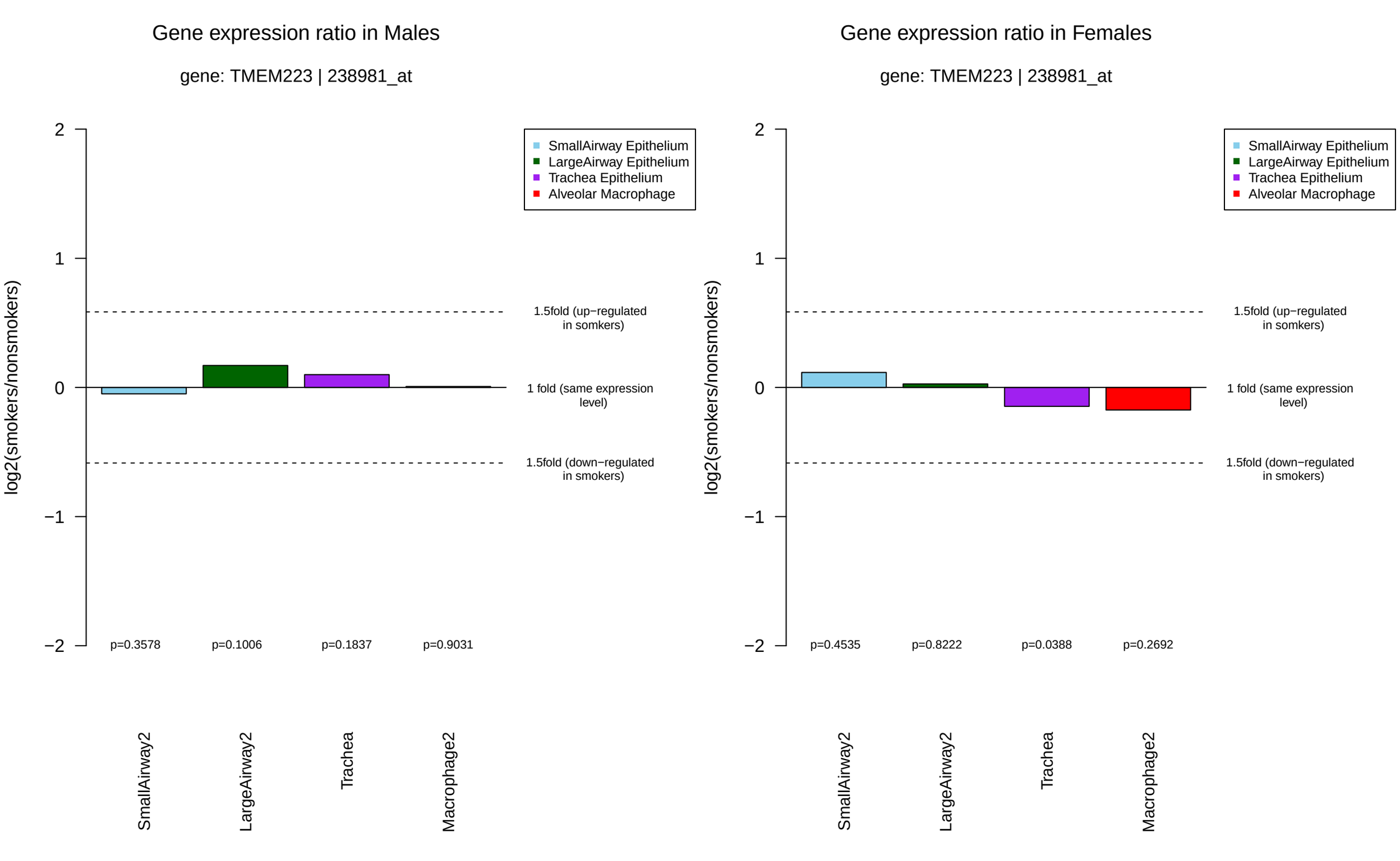
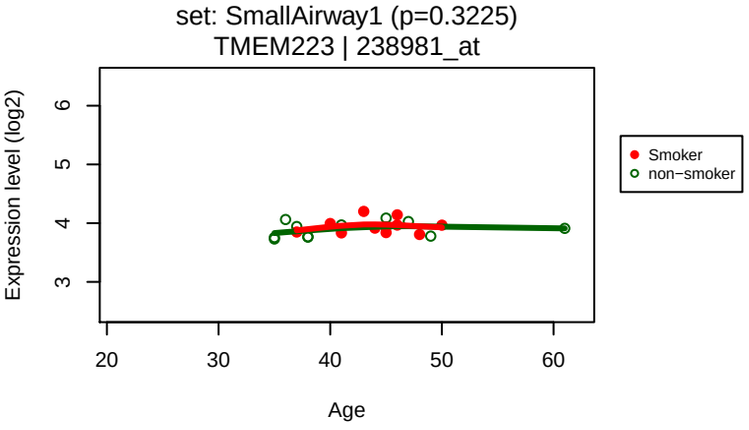
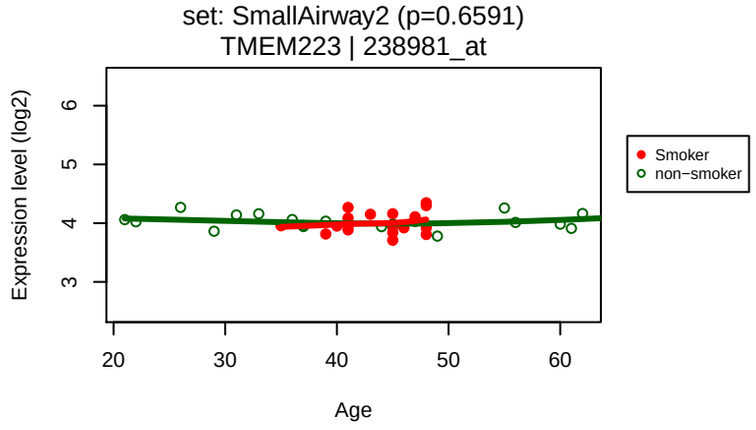
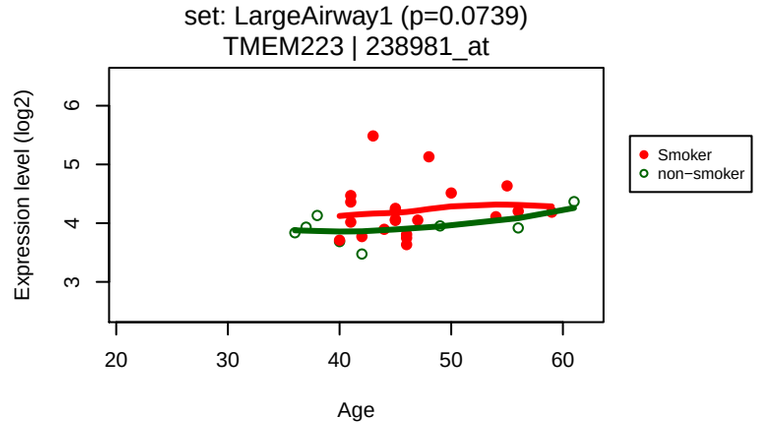

Gene expression of smokers and nonsmokers
gene: TMEM223 | 238981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3225
p=0.6591
p=0.4268
p=0.0739
p=0.0899
p=0.3775
p=0.2707
p=0.3947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

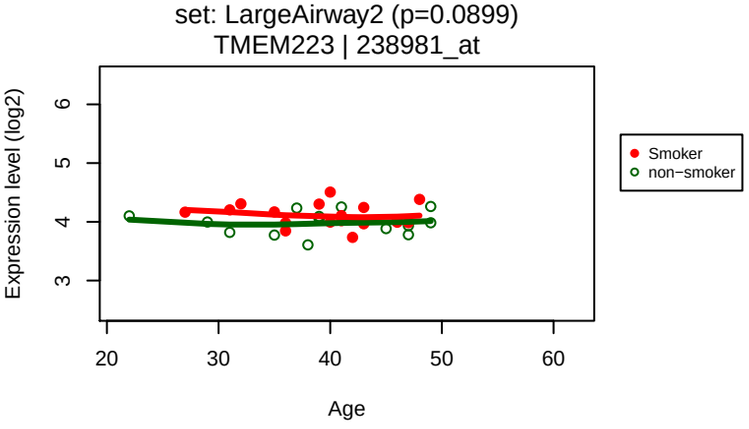

Gene expression ratio (smokers/nonsmokers)
gene: TMEM223 | 238981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3225
p=0.6591
p=0.4268
p=0.0739
p=0.0899
p=0.3775
p=0.2707
p=0.3947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

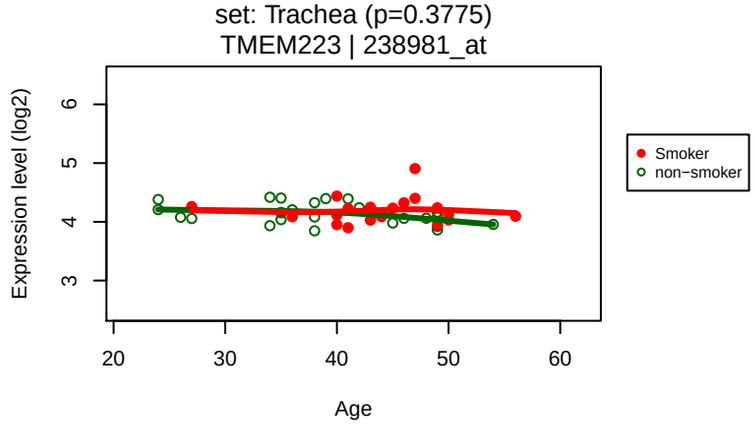

Gene expression ratio in Males
gene: TMEM223 | 238981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3578
p=0.1006
p=0.1837
p=0.9031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM223 | 238981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4535
p=0.8222
p=0.0388
p=0.2692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

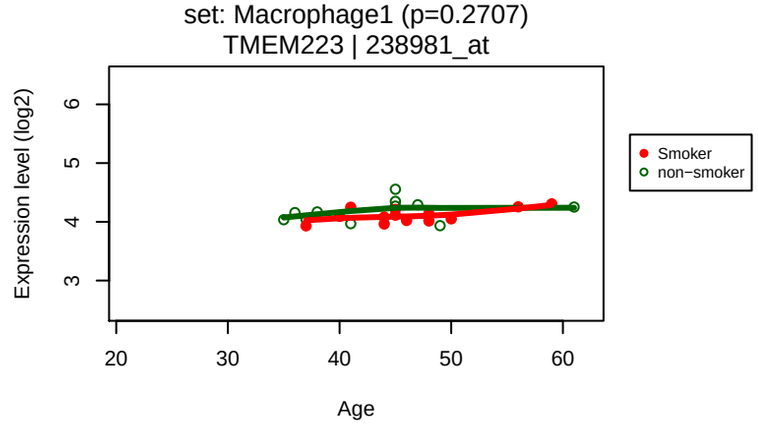

set: SmallAirway1 (p=0.3225)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker

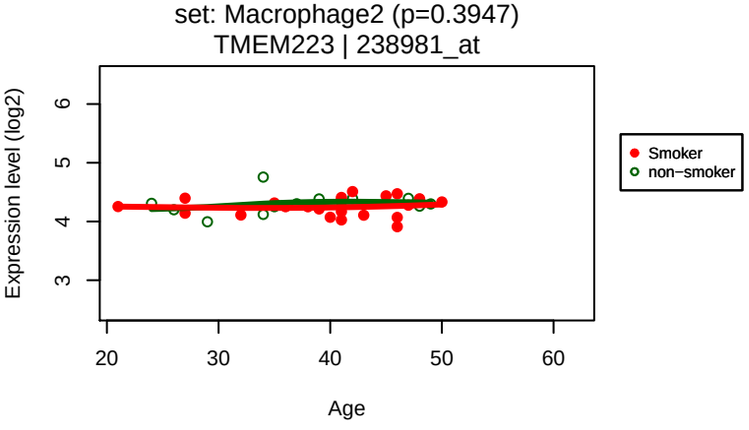

set: SmallAirway2 (p=0.6591)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker

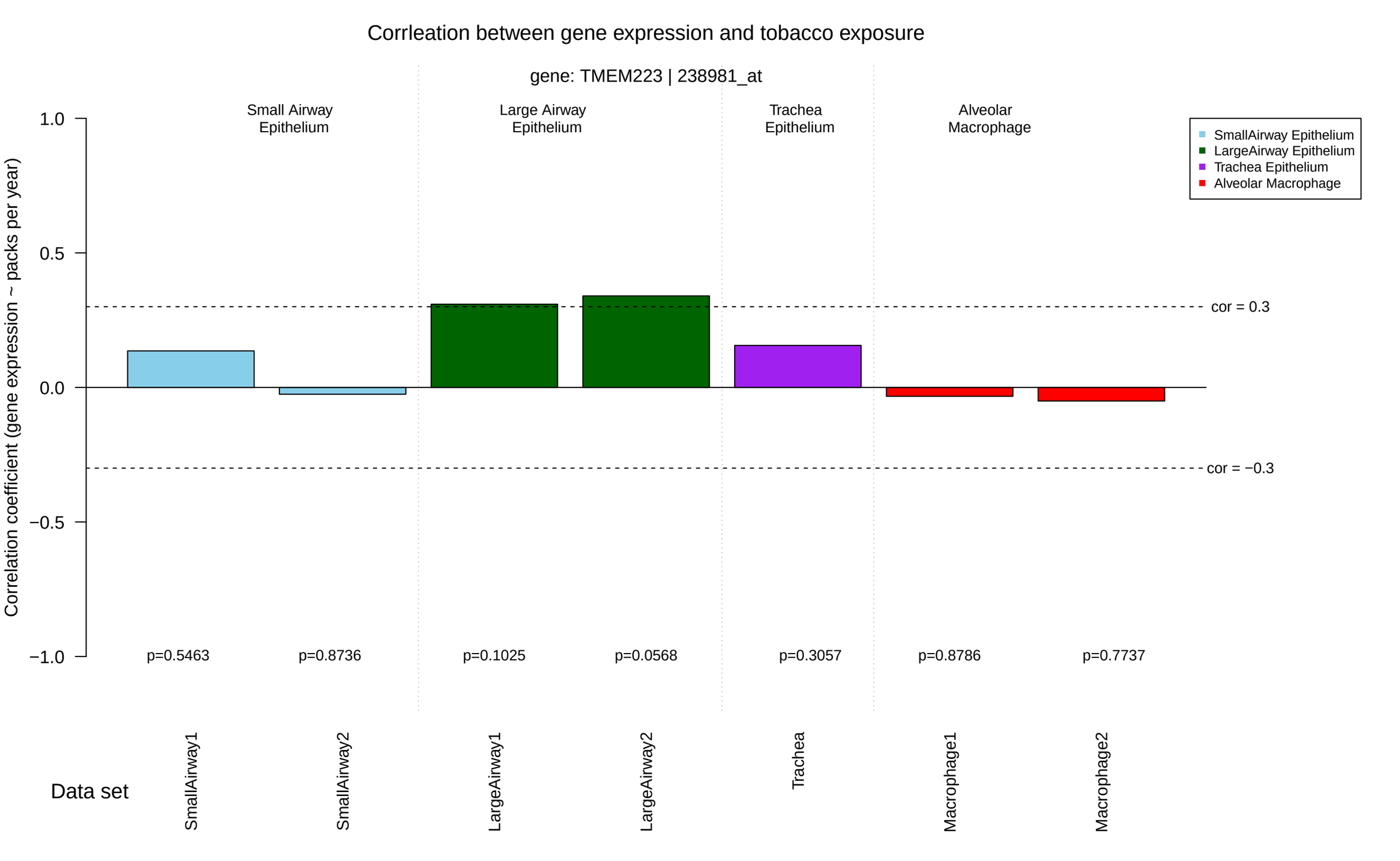

set: LargeAirway1 (p=0.0739)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0899)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3775)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2707)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3947)
TMEM223 | 238981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM223 | 238981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5463
p=0.8736
p=0.1025
p=0.0568
p=0.3057
p=0.8786
p=0.7737
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2